MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746262
COMPUTER -1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-G-48411 comprise the complete requirements for Computer-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the data of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746262	Circuit Card Assembly, Computer-1

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Identification.- This assembly is one (1) of six (6) single ended printed circuit boards consisting multiplier, operational amplifiers diodes and sin/cos generators necessary to provide mathematical manipulation of various parameters for the generation of computer flight paths.

3.1.1 Major component list.- Ths components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746262.

3.2 Characteristics.

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3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +10 VDC
b. -10 VDC
c. +15 VDC
d. -15 VDC

3.2.1.2 Multiplier Test.- The multiplier shall be capable of providing a DC output voltage proportional to the algebraic product of difference of two voltages when the input voltages are within the limits of ± 10 VDC.

3.2.1.3 Sine/Cosine Generator Test.- The generator shall be capable of providing a DC voltage output proportional to the sine/cosine on an input voltage where ± 10 VDC represent ± 90 degrees of input angle.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on Drawing 11746262 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 11746262 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

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4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power Supply (PS1 & PS2).- The power supplies shall be 15 ± 0.001 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS3 & PS4).- The power supply shall be 20 ± 0.01 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Dummy Loads.- The dummy loads shall be 10K, 1/4 W, 1% resistors.

4.2.1.4 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

4.2.2.2 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3 Multiplier Test.- When specific voltage values of between I 10 VDC are applied to the inputs of the multiplier the corresponding output shall be as shown in Table I.

TABLE I

	Input(VDC)				Output(VDC)
Pins	22	24	29	30	25
	0	0	0	0	0.000 ± 0.001 obtained by adjusting R23
	1	5	5	1	-1.6 ± 0.1
	10	1	-10	-1	-8.1 ± 0.1
	10	5	10	5	+2.5 ± 0.1
	-10	-1	-10	-1	8.1 ± 0.1

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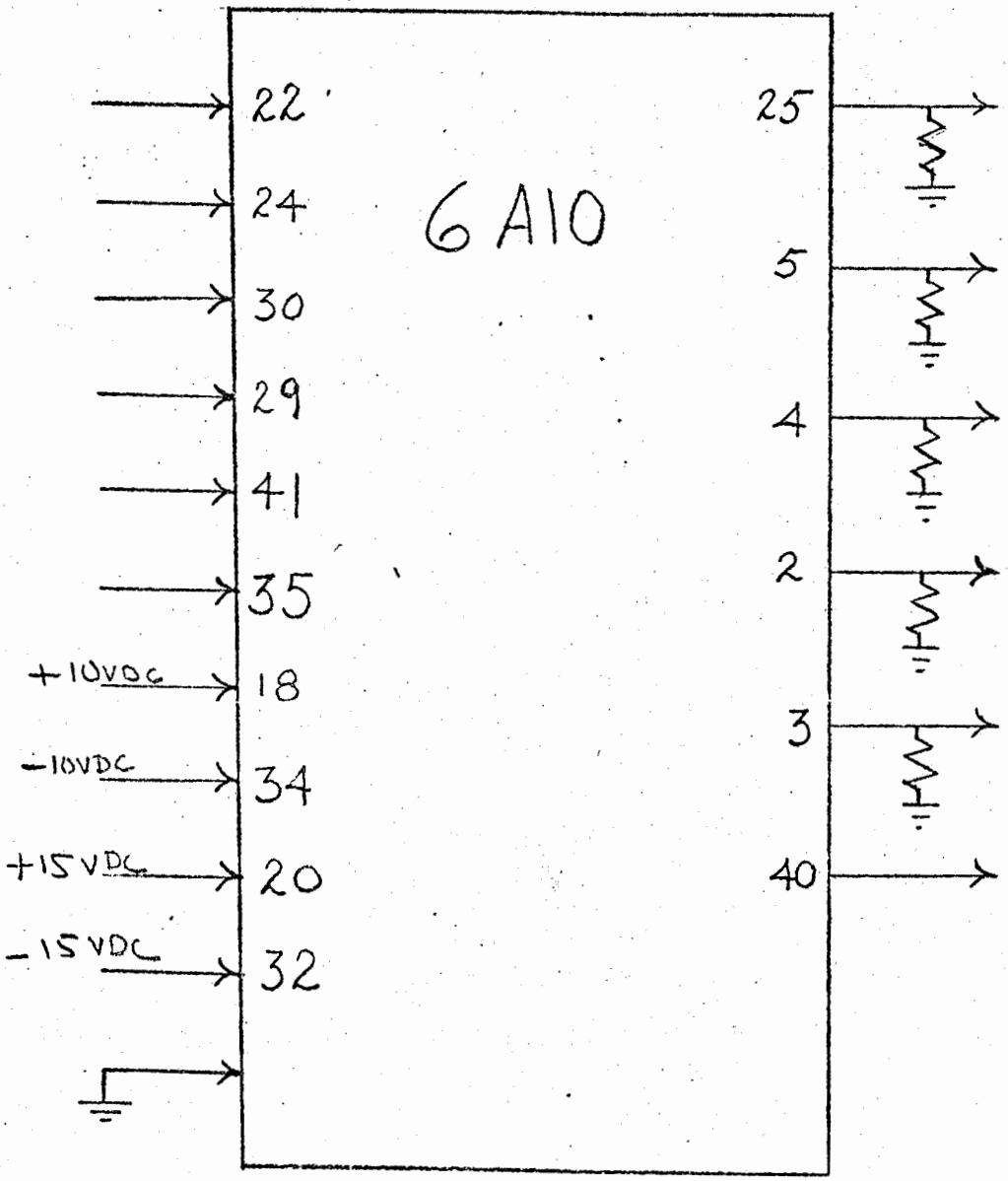


FIGURE 1

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NOTE: The output tolerances during testing at temperature extremes shall be ± 0.30 VDC.

4.2.2.4 Sine/Cosine Generator Test.- When specific voltage values of ± 10 VDC are applied to the input pin 41 and 35 of the sin/cos generator network the corresponding outputs shall be on the pins as shown in Table II.

TABLE II

	Input (VDC ± 1 mV)		Output (VDC ± 1 mV)			
Pins	41	35	5	4	3	2
	0.000	0.000	0.000*	0	10.000	0
	5.000	0.000	10.000	-	0.000*	0
	10.000	0.000	0.000	-	-10.000	-
	7.000	0.000	8.090	-	5.880	-
	2.000	0.000	5.880	-	8.090	-
	-2.000	0.000	-5.880	-	-8.090	-
	-5.000	0.000	-10.000	-	0.000	-
	-7.000	0.000	8.090	-	-5.880	-

* If output is not 0.0 ± 1 mV on pin 5 and 3, obtain same by adjusting R17 or R18 prior to proceeding.

NOTE: The output tolerances during testing at temperature extremes shall be ± 0.400 VDC.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746262.

6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

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Preparing Activity:

Army - MU

Project No. 1240-A605

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